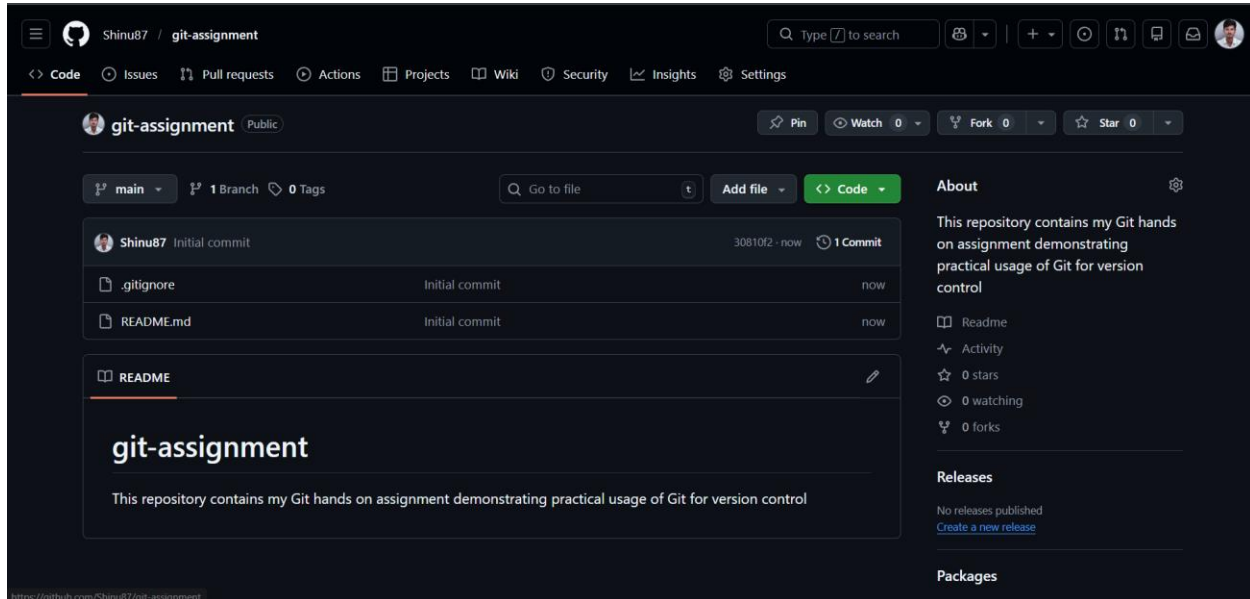


# Topic : Git Commands and Workflows

## 1. Create a New Repository

A new repository was created on Github to show git operation

Git Repo Link : <https://github.com/Shinu87/git-assignment.git>



## 2.Create a New File

First, the repository created on github was cloned to the local system.

Then a new file named file.txt was created using the touch command and content was added to it.

```
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo$ git clone https://github.com/Shinu87/git-assignment.git
Cloning into 'git-assignment'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), done.
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo$ cd git-assignment
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ touch file.txt
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ echo "This is my git assignment file" >> file.txt
```

After creating the file, the status of the repository was checked using the git status command.

The file appeared as an untracked file.

Then file was added to staging area and then committed

## Topic : Git Commands and Workflows

```
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
       file.txt

nothing added to commit but untracked files present (use "git add" to track)
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git add .
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
       new file:   file.txt

shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git commit -m "added file.txt"
[main 4250352] added file.txt
 1 file changed, 1 insertion(+)
 create mode 100644 file.txt
```

### 3) Create a Branch / 4) Make Changes in the New Branch

A new branch is created for new feature i.e. feature1 without affecting main branch.

A change was made in file.txt and committed to the branch.

```
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git checkout -b feature1
Switched to a new branch 'feature1'
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ echo "Change made in feature1 branch" >> file.txt
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git add .
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git commit -m "updated file in feature1 branch"
[feature1 65f19a5] updated file in feature1 branch
 1 file changed, 1 insertion(+)
```

### 5) Create Another Branch

Another branch with name feature2 was created to show working with multiple branches

```
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git checkout -b feature2
Switched to a new branch 'feature2'
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ echo "Change made in feature2 branch" >> file.txt
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git add .
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git commit -m "updated file in feature2 branch"
[feature2 6a3cdb4] updated file in feature2 branch
 1 file changed, 1 insertion(+)

shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git branch
  feature1
* feature2
   main
```

## Topic : Git Commands and Workflows

### 6) Merge Branches

First switched to the main branch

Then merged the branches

```
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git merge feature1
Updating 4250352..65f19a5
Fast-forward
 file.txt | 1 +
 1 file changed, 1 insertion(+)
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git merge feature2
Updating 65f19a5..6a3cdb4
Fast-forward
 file.txt | 1 +
 1 file changed, 1 insertion(+)
```

### 7) Create an .env file, add the email to it, add that .env in .gitignore

A .env file was created to store the email

To prevent this file from being tracked it was added to gitignore

Then gitignore file was then staged and committed

```
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ touch .env
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ echo "Email:shrinvasmore51@gmail.com"> .env
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ cat .env
Email:shrinvasmore51@gmail.com
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ echo ".env" > .gitignore
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ cat .gitignore
.env
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git status
On branch main
Your branch is ahead of 'origin/main' by 3 commits.
(use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   .gitignore

no changes added to commit (use "git add" and/or "git commit -a")
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git add .gitignore
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git commit -m "Added .env file"
[main a40fffa] Added .env file
 1 file changed, 1 insertion(+), 207 deletions(-)
rewrite .gitignore (99%)
```

## Topic : Git Commands and Workflows

```
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git push origin main
Username for 'https://github.com': Shinu87
Password for 'https://Shinu87@github.com':
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 16 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (12/12), 1.11 KiB | 13.00 KiB/s, done.
Total 12 (delta 3), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (3/3), done.
To https://github.com/Shinu87/git-assignment.git
30810f2..a40fffa main -> main
```

The screenshot shows the GitHub web interface for a repository named 'git-assignment' owned by 'Shinu87'. The repository is public and has 0 stars, 0 forks, and 0 watchers. The main branch is 'main'. The commit history shows 5 commits. The most recent commit is 'Added .env file' by 'Shinu87' 19 minutes ago, with commit hash 'a40fffa'. The repository contains files: '.gitignore', 'README.md', and 'file.txt'. The README file is selected and shows the title 'git-assignment' and the description 'This repository contains my Git hands on assignment demonstrating practical usage of Git for version control'.

### 8) Pull Changes from Remote

The latest changes from the remote repository were pulled to local system

```
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git pull origin main
From https://github.com/Shinu87/git-assignment
* branch      main      -> FETCH_HEAD
Already up to date.
```

## Topic : Git Commands and Workflows

### 9) View Git Log and History

The commit history of the repository was viewed using the git log command

```
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git log
commit a40fffa797e6c5d62a936c8e5dc3ceb6bf159a1a (HEAD -> main, origin/main, origin/HEAD)
Author: Shinu87 <shrinvasmore51@gmail.com>
Date: Tue Mar 10 09:46:50 2026 +0530

    Added .env file

commit 6a3cdb41817f8a36138074e3a4f0891ffb8dcf3f (feature2)
Author: Shinu87 <shrinvasmore51@gmail.com>
Date: Tue Mar 10 09:36:56 2026 +0530

    updated file in feature2 branch

commit 65f19a51074ca8b2eeea4bc6ceb7f2464138a280 (feature1)
Author: Shinu87 <shrinvasmore51@gmail.com>
Date: Tue Mar 10 09:34:07 2026 +0530

    updated file in feature1 branch

commit 42503521acf17f8d0484c1ec6d63abe42797e061
Author: Shinu87 <shrinvasmore51@gmail.com>
Date: Tue Mar 10 09:31:14 2026 +0530

    added file.txt

commit a40fffa797e6c5d62a936c8e5dc3ceb6bf159a1a (HEAD -> main, origin/main, origin/HEAD)
Author: Shinu87 <shrinvasmore51@gmail.com>
Date: Tue Mar 10 09:46:50 2026 +0530

    Added .env file

commit 6a3cdb41817f8a36138074e3a4f0891ffb8dcf3f (feature2)
Author: Shinu87 <shrinvasmore51@gmail.com>
Date: Tue Mar 10 09:36:56 2026 +0530

    updated file in feature2 branch

commit 65f19a51074ca8b2eeea4bc6ceb7f2464138a280 (feature1)
Author: Shinu87 <shrinvasmore51@gmail.com>
Date: Tue Mar 10 09:34:07 2026 +0530

    updated file in feature1 branch

commit 42503521acf17f8d0484c1ec6d63abe42797e061
Author: Shinu87 <shrinvasmore51@gmail.com>
Date: Tue Mar 10 09:31:14 2026 +0530

    added file.txt
```

## Topic : Git Commands and Workflows

```
commit 30810f2747f835bbe8b613d226f36bd8cacfd8
commit a40fffa797e6c5d62a936c8e5dc3ceb6bf159a1a (HEAD -> main, origin/main, origin/HEAD)
Author: Shinu87 <shrinvasmore51@gmail.com>
Date: Tue Mar 10 09:46:50 2026 +0530

    Added .env file

commit 6a3cdb41817f8a36138074e3a4f0891ffb8dcf3f (feature2)
Author: Shinu87 <shrinvasmore51@gmail.com>
Date: Tue Mar 10 09:36:56 2026 +0530

    updated file in feature2 branch

commit 65f19a51074ca8b2eeea4bc6eb7f2464138a280 (feature1)
Author: Shinu87 <shrinvasmore51@gmail.com>
Date: Tue Mar 10 09:34:07 2026 +0530

    updated file in feature1 branch

commit 42503521acf17f8d0484c1ec6d63abe42797e061
Author: Shinu87 <shrinvasmore51@gmail.com>
Date: Tue Mar 10 09:31:14 2026 +0530

    added file.txt

commit 30810f2747f835bbe8b613d226f36bd8cacfd8
Author: MORE SHRINIVAS BALAJI <142227902+Shinu87@users.noreply.github.com>
:...skipping...

commit a40fffa797e6c5d62a936c8e5dc3ceb6bf159a1a (HEAD -> main, origin/main, origin/HEAD)
Author: Shinu87 <shrinvasmore51@gmail.com>
Date: Tue Mar 10 09:46:50 2026 +0530

    Added .env file

commit 6a3cdb41817f8a36138074e3a4f0891ffb8dcf3f (feature2)
Author: Shinu87 <shrinvasmore51@gmail.com>
Date: Tue Mar 10 09:36:56 2026 +0530

    updated file in feature2 branch

commit 65f19a51074ca8b2eeea4bc6eb7f2464138a280 (feature1)
Author: Shinu87 <shrinvasmore51@gmail.com>
Date: Tue Mar 10 09:34:07 2026 +0530

    updated file in feature1 branch

commit 42503521acf17f8d0484c1ec6d63abe42797e061
Author: Shinu87 <shrinvasmore51@gmail.com>
Date: Tue Mar 10 09:31:14 2026 +0530

    added file.txt

commit 30810f2747f835bbe8b613d226f36bd8cacfd8
Author: MORE SHRINIVAS BALAJI <142227902+Shinu87@users.noreply.github.com>
Date: Tue Mar 10 09:18:52 2026 +0530
:...skipping...
```

## Topic : Git Commands and Workflows

```
commit a40fffa797e6c5d62a936c8e5dc3ceb6bf159a1a (HEAD -> main, origin/main, origin/HEAD)
Author: Shinu87 <shrinivasmore51@gmail.com>
Date: Tue Mar 10 09:46:50 2026 +0530
```

```
Added .env file
```

```
commit 6a3cdb41817f8a36138074e3a4f0891ffb8dcf3f (feature2)
Author: Shinu87 <shrinivasmore51@gmail.com>
Date: Tue Mar 10 09:36:56 2026 +0530
```

```
updated file in feature2 branch
```

```
commit 65f19a51074ca8b2eeea4bc6ceb7f2464138a280 (feature1)
Author: Shinu87 <shrinivasmore51@gmail.com>
Date: Tue Mar 10 09:34:07 2026 +0530
```

```
updated file in feature1 branch
```

```
commit 42503521acf17f8d0484c1ec6d63abe42797e061
Author: Shinu87 <shrinivasmore51@gmail.com>
Date: Tue Mar 10 09:31:14 2026 +0530
```

```
added file.txt
```

```
commit 30810f2747f835bbe8b613d226f36bd8cacfd8ae8
Author: MORE SHRINIVAS BALAJI <142227902+Shinu87@users.noreply.github.com>
Date: Tue Mar 10 09:18:52 2026 +0530
```

```
:...skipping...
```

```
commit a40fffa797e6c5d62a936c8e5dc3ceb6bf159a1a (HEAD -> main, origin/main, origin/HEAD)
Author: Shinu87 <shrinivasmore51@gmail.com>
Date: Tue Mar 10 09:46:50 2026 +0530
```

```
Added .env file
```

```
commit 6a3cdb41817f8a36138074e3a4f0891ffb8dcf3f (feature2)
Author: Shinu87 <shrinivasmore51@gmail.com>
Date: Tue Mar 10 09:36:56 2026 +0530
```

```
updated file in feature2 branch
```

```
commit 65f19a51074ca8b2eeea4bc6ceb7f2464138a280 (feature1)
Author: Shinu87 <shrinivasmore51@gmail.com>
Date: Tue Mar 10 09:34:07 2026 +0530
```

```
updated file in feature1 branch
```

```
commit 42503521acf17f8d0484c1ec6d63abe42797e061
Author: Shinu87 <shrinivasmore51@gmail.com>
Date: Tue Mar 10 09:31:14 2026 +0530
```

```
added file.txt
```

```
commit 30810f2747f835bbe8b613d226f36bd8cacfd8ae8
Author: MORE SHRINIVAS BALAJI <142227902+Shinu87@users.noreply.github.com>
Date: Tue Mar 10 09:18:52 2026 +0530
```

```
Initial commit
```

```
:...skipping...
```

## Topic : Git Commands and Workflows

```
commit a40fffa797e6c5d62a936c8e5dc3ceb6bf159a1a (HEAD -> main, origin/main, origin/HEAD)
Author: Shinu87 <shrinvasmore51@gmail.com>
Date: Tue Mar 10 09:46:50 2026 +0530

    Added .env file

commit 6a3cdb41817f8a36138074e3a4f0891ffb8dcf3f (feature2)
Author: Shinu87 <shrinvasmore51@gmail.com>
Date: Tue Mar 10 09:36:56 2026 +0530

    updated file in feature2 branch

commit 65f19a51074ca8b2eeea4bc6ceb7f2464138a280 (feature1)
Author: Shinu87 <shrinvasmore51@gmail.com>
Date: Tue Mar 10 09:34:07 2026 +0530

    updated file in feature1 branch

commit 42503521acf17f8d0484c1ec6d63abe42797e061
Author: Shinu87 <shrinvasmore51@gmail.com>
Date: Tue Mar 10 09:31:14 2026 +0530

    added file.txt

commit 30810f2747f835bbe8b613d226f36bd8cacfd8e8
Author: MORE SHRINIVAS BALAJI <142227902+Shinu87@users.noreply.github.com>
Date: Tue Mar 10 09:18:52 2026 +0530

    Initial commit
~
~
~
~
~
(END)
```

To get a visual representation of the commit history the following command was used

```
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git log --oneline --graph --all
* a40fffa (HEAD -> main, origin/main, origin/HEAD) Added .env file
* 6a3cdb4 (feature2) updated file in feature2 branch
* 65f19a5 (feature1) updated file in feature1 branch
* 4250352 added file.txt
* 30810f2 Initial commit
```



## Topic : Git Commands and Workflows

### 10) Git Status and Staging

First I modified existing file.txt to demonstrate this step

Git status command was used to check the current status of repository it shows file.txt as modified.

Then file was added to the staging area.

Finally the changes were committed.

```
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   file.txt

no changes added to commit (use "git add" and/or "git commit -a")
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git add .
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   file.txt

shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git commit -m "git status and staging"
[main 74f9233] git status and staging
1 file changed, 1 insertion(+)
```

### 11) Revert and Reset

To undo the most recent commit safely git revert command was used

```
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git revert HEAD
[main 0ec9ce1] Revert "git status and staging"
1 file changed, 1 deletion(-)
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ 

GNU nano 6.2 /mnt/c/Users/shrin/Desktop/GitDemo/git-assignment/.git/COMMIT_EDITMSG
Revert "git status and staging"

This reverts commit 74f9233af150ae556876a276d6cd1add7185edec.

# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
#
# On branch main
# Your branch is ahead of 'origin/main' by 1 commit.
#   (use "git push" to publish your local commits)
#
# Changes to be committed:
#       modified:   file.txt
#
#
```

## Topic : Git Commands and Workflows

The commit history was verified

```
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git log
commit 0ec9ce10011ca69a34261fbd4d4d152e42f17a3b (HEAD -> main)
Author: Shinu87 <shrinivasmore51@gmail.com>
Date: Tue Mar 10 10:27:27 2026 +0530

    Revert "git status and staging"

    This reverts commit 74f9233af150ae556876a276d6cd1add7185edec.

commit 74f9233af150ae556876a276d6cd1add7185edec
Author: Shinu87 <shrinivasmore51@gmail.com>
Date: Tue Mar 10 10:26:09 2026 +0530

    git status and staging
```

### Additional Task: Merge Conflict and Resolution

Switched to feature1 branch and made changes in file.txt.

Switched to main branch and modified the file.txt.

When tried to merge feature1 into main caused a merge conflict.

I have manually edited to resolve the conflict.

And verified commit history.

```
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git checkout feature1
Switched to branch 'feature1'
t add file.txt
t add file.txt
t add file.txt
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git commit -m "change in feature1 branch"
[feature1 27fcf54] change in feature1 branch
1 file changed, 1 insertion(+), 2 deletions(-)
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 2 commits.
(use "git push" to publish your local commits)
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ echo "Hello from main branch" > file.txt
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git add file.txt
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git commit -m "change in main branch"
[main d8f319f] change in main branch
1 file changed, 1 insertion(+), 3 deletions(-)
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git merge feature1
Auto-merging file.txt
CONFLICT (content): Merge conflict in file.txt
Automatic merge failed; fix conflicts and then commit the result.
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git add file.txt
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git commit -m "conflict resolved"
[main 7c56265] conflict resolved
shinu@Shinu:/mnt/c/Users/shrin/Desktop/GitDemo/git-assignment$ git log --oneline --graph
* 7c56265 (HEAD -> main) conflict resolved
|
| * 27fcf54 (feature1) change in feature1 branch
| * d8f319f change in main branch
| * 0ec9ce1 Revert "git status and staging"
| * 74f9233 git status and staging
| * a40fffa (origin/main, origin/HEAD) Added .env file
| * 6a3cdb4 (feature2) updated file in feature2 branch
|/
* 65f19a5 updated file in feature1 branch
* 4250352 added file.txt
* 30810f2 Initial commit
```

Accept Current Change | Accept Incoming Change | Accept Both Changes | Compare Changes

<<<<<< HEAD (Current Change)

Hello from main branch

=====

Hello from feature1 branch

>>>>>> feature1 (Incoming Change)